

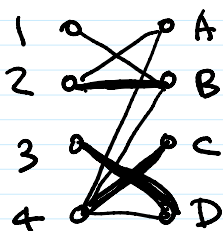
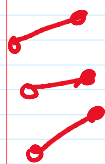
OPTIMIZATION PROBLEMS:

Matchings, Flows, & LPs

Problem Given unweighted bipartite graph $G=(V,E)$,
 $|V|=n$, $|E|=m$,
 $(1 \leq n \leq n^2)$

find matching with max size

Subgraph where every vertex $\deg \leq 1$.



$\{2B, 3D, 4C\}$ ← aug path $1B2A$

$\{1B, 2A, 3D, 4C\}$ max 4.

idea - brute force $\Omega(n!)$

greedy, not work

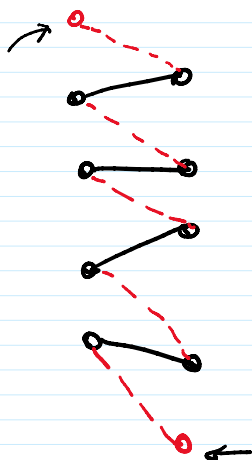
DP?

D&C?

"Hungarian" Method (König-Egerváry '55 / Edmonds '65)

idea - local improvement

(start with feasible sol'n, modify & improve, ...)



"alternating path"
augmenting

Alg'm:

$M = \emptyset$

repeat: {

find aug path P for M

if not exist, return M

↖

find aug path P for M

if not exist, return M

modify M by P . $\leftarrow M = M \oplus P$

(Facts a matching M is optimal $\Leftrightarrow$ it has no augmenting path)
 can find an augmenting path in $O(m)$ time by BFS/DFS
 or decide none exists

iterations $\leq n/2 \leftarrow$ because $|M|$ increments per iteration
 $\Rightarrow$ total time $O(mn)$

⋮

Def. Let M be a matching.

An augmenting path, P for M is a path

$v_1 v_2 v_3 \dots v_k$ in G

s.t. $v_1 v_2 \notin M$, $v_2 v_3 \in M$,

$v_3 v_4 \notin M$, $v_4 v_5 \in M$,

(even)

$\dots$, $v_{k-1} v_k \notin M$.

and v_1, v_k are not covered by M
 ("free")

$\Rightarrow M \oplus P$ is a matching
 with $|M \oplus P| = |M| + 1$.

$\oplus$
 symmetric diff $S \oplus T = S \cup T - (S \cap T)$

Correctness:

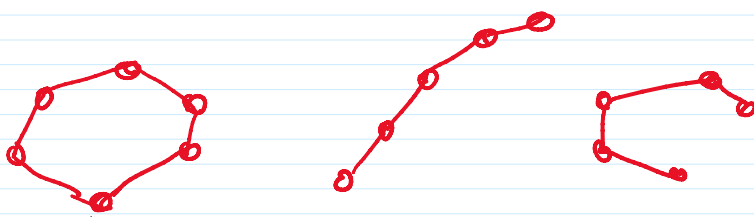
Thm A matching M is optimal $\Leftrightarrow$ it has no augmenting path.

$P \rightarrow Q$
 $Q \rightarrow P$ sum $\rightarrow P \rightarrow Q$

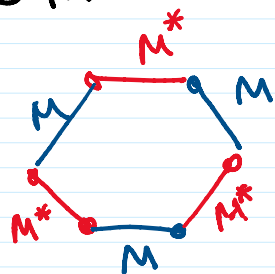
Lemma If M and M^* are matchings with $|M| < |M^*|$,
 then $\exists \geq |M^*| - |M|$ aug paths for M .
 vertex-disjoint

Pf: $M \oplus M^*$ is a subgraph with $\max \deg \leq 2$

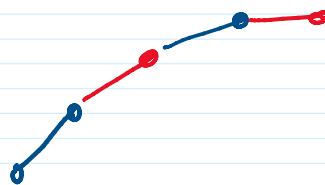
Fact: an undir graph with $\max \deg \leq 2$
 must be vertex-disjoint union of paths & cycles.



$M \oplus M^*$:



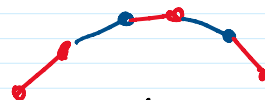
cycle of
even length



path of
even length



blue-
odd path



red-
odd path

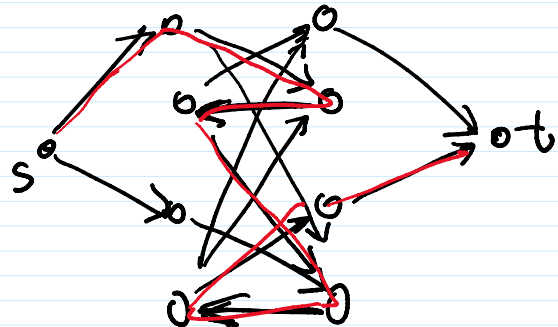
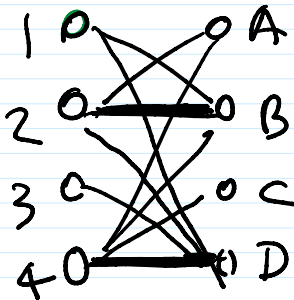
$$\begin{aligned}
 |M^*| - |M| &= \# \text{ red-odd paths} - \# \text{ blue-odd paths} \\
 &\leq \# \text{ red-odd paths} \\
 &\leq \# \text{ aug paths} \quad \square
 \end{aligned}$$

Pf of Thm: $\exists$ aug path $\Rightarrow$ not optimal: easy

not opt $\Rightarrow \exists$ aug path: by lemma. $\square$

Analysis:

How to find aug path?



form a new dir graph G_M

for u on left, v on right,

if $uv \in M$, add (v, u) to G_M

if $uv \in G \setminus M$, add (u, v) to G_M

if u is free, add (s, u) to G_M

if v is free, add (v, t) to G_M .

$O(m)$
time

run BFS/DFS on G_M

total run time $O(mn)$

Hopcroft & Karp's Alg ('73): Sketch

idea 1 - always pick shortest augmenting path

idea 2 - augment multiple ^{vertex-disjoint} shortest augment paths
all in $O(m)$ time

Fact length of shortest aug paths increase
per iteration

$\Rightarrow$ after $\geq \sqrt{n}$ iterations, path lengths $\geq \sqrt{n}$

$\Rightarrow \leq \sqrt{n}$ vertex-disj. aug paths
by lemma $\Rightarrow |M^*| - |M| \leq \sqrt{n}$

$\Rightarrow |M| \geq |M^*| - \sqrt{n}$

$\Rightarrow$ finish in $\leq \sqrt{n}$ iterations

$\Rightarrow \boxed{O(m\sqrt{n})}$ time

...

$O(m^{1+\epsilon})$ time [van den Brand, Lee,
Nanongkai, Peng,
Saranurak, Sidford, Song,
Wang '20]

Extensions:

max unweighted non-bipartite

Edmonds '65 $O(mn^2)$

↓

Micali, Vazirani '80 $O(m\sqrt{n})$

max weighted bipartite

max weighted non-bipartite

min-cost perfect matching

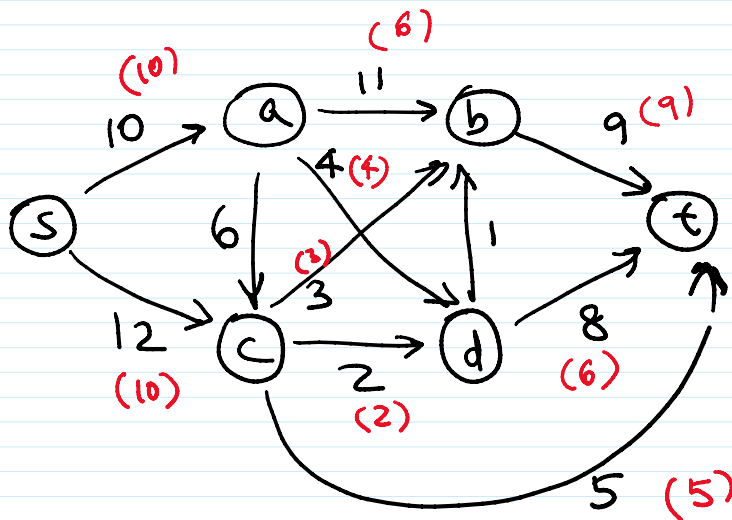
Problem max flow

"source"
(in-deg 0)

"destination"
(out-deg 0)

Given dir graph $G = (V, E)$, $s, t \in V$,
where each $e \in E$ has a capacity $c(e) > 0$,

find flow from s to t with max value



$$\begin{aligned} 10 + 10 \\ &= 9 + 6 + 5 \\ &= 20 \end{aligned}$$